

- Q.28 Distinguish between structure and union.
- Q.29 Write a program in C to calculate factorial of a given number using loops.
- Q.30 What is an array? Give example of 1-D and 2-D arrays.
- Q.31 How to declare local and Global variables in a program?
- Q.32 Explain pointers . How are they declared.
- Q.33 Explain various data types available in 'C' language.
- Q.34 What is the address operator in C?
- Q.35 Write a program in C to swap two numbers.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What is switch statement in C? Explain its usage with an example.
- Q.37 Explain the concept of unions in C programming with appropriate examples. Also, write their advantages, and limitations compared to other data structures.
- Q.38 Define function. How do you pass arguments to a function in call by value.

No. of Printed Pages : 4 180836/170836/082435
Roll No.

3rd Sem / Comp. / Mechtronics Subject:- Programming in C

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is typically used to represent the logical flow of a program before actual coding begins in C?
- a) Flor charts b) Header Files
c) Variables d) Constants
- Q.2 Which of the following is a correct variable name,
- a) int salary_23; b) int salary 23;
c) int salary-23; d) int salary 2_3;
- Q.3 How to initialize an array?
- a) int a(3)=[1,2,3]; b) int a[3]=[1,2,3,4],
c) int a[3]={1,2,3}; d) None of the above
- Q.4 Which of the following is an example of iteration in C?
- a) for
b) while
c) do-while
d) all of the mentioned above

- Q.5 C was invented by?
- a) Tim berners Lee b) Denis Ritchie
c) Bjarne Stroustrup d) Rasmus Lerdorf
- Q.6 The `#include <stdio.h>` is a _____.
- a) Inclusion directive
b) File inclusion directive
c) Preprocessor directive
d) None of the above
- Q.7 How do you access the third element of an array using pointer arithmetic?
- a) `*(arr+2)` b) `*(arr + 3)`
c) `arr[3]` d) `*arr[3]`
- Q.8 How do you access members of a structure or union pointer in C?
- a) Using the arrow operator (`->`)
b) Using the dot operator (`.`)
c) Using the asterisk operator (`*`)
d) None of the above
- Q.9 Which data type is used to store characters in C?
- a) `char` b) `int`
c) `float` d) `double`
- Q.10 Which of the following is a keyword in C programming language?
- a) `loop` b) `print`
c) `main` d) `if`

(2) 180836/170836/082435

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is relational operator?
- Q.12 How to add single line comments in C?
- Q.13 How to calculate size of an array in 1-D?
- Q.14 C language is case sensitive language _____ (True/False)
- Q.15 Define keyword.
- Q.16 Name any two header files used in C.
- Q.17 How to call a function in C?
- Q.18 How to represent condition in flowchart?
- Q.19 Break statement is used for _____
- Q.20 Output of `15%2` in C programming _____

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are the steps involved in the development of a program?
- Q.22 What methods are commonly used for program debugging in C?
- Q.23 Define constants and variables in the context of programming.
- Q.24 Explain arithmetic operators in C programming.
- Q.25 What is for loop? Explain with an example.
- Q.26 What are the decision making statements in C?
- Q.27 How can you compare and concentrate two strings? Explain with examples.

(3) 180836/170836/082435